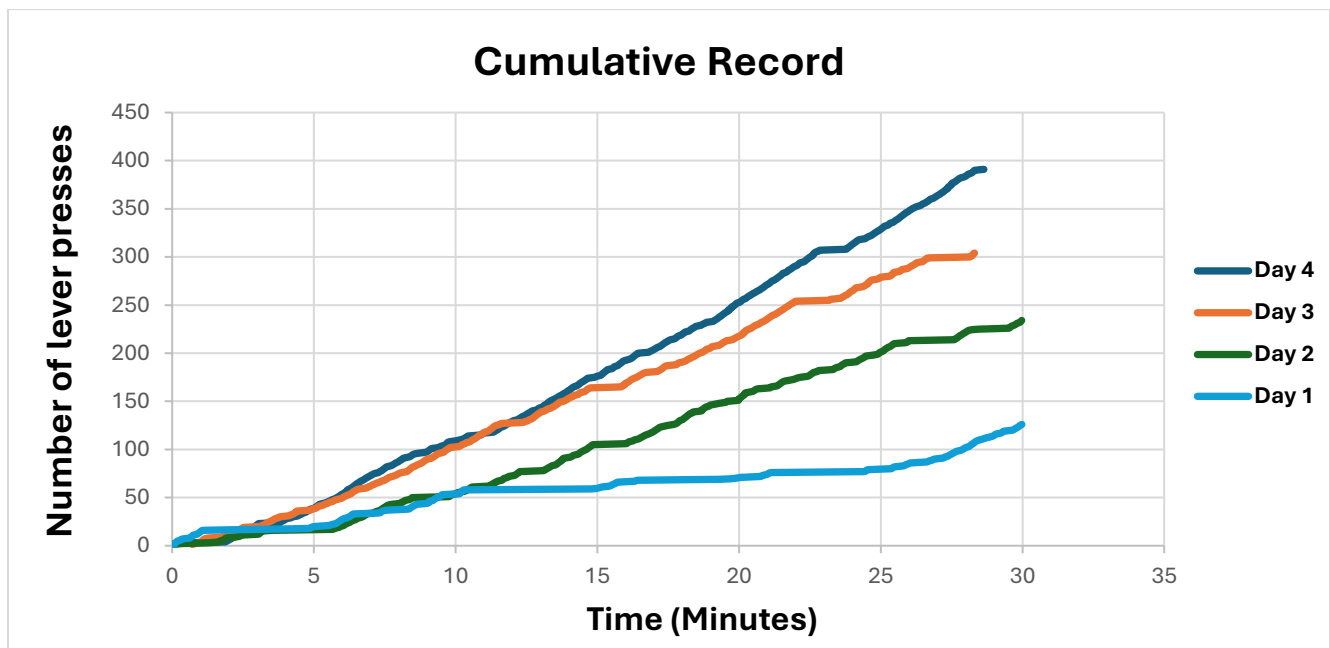


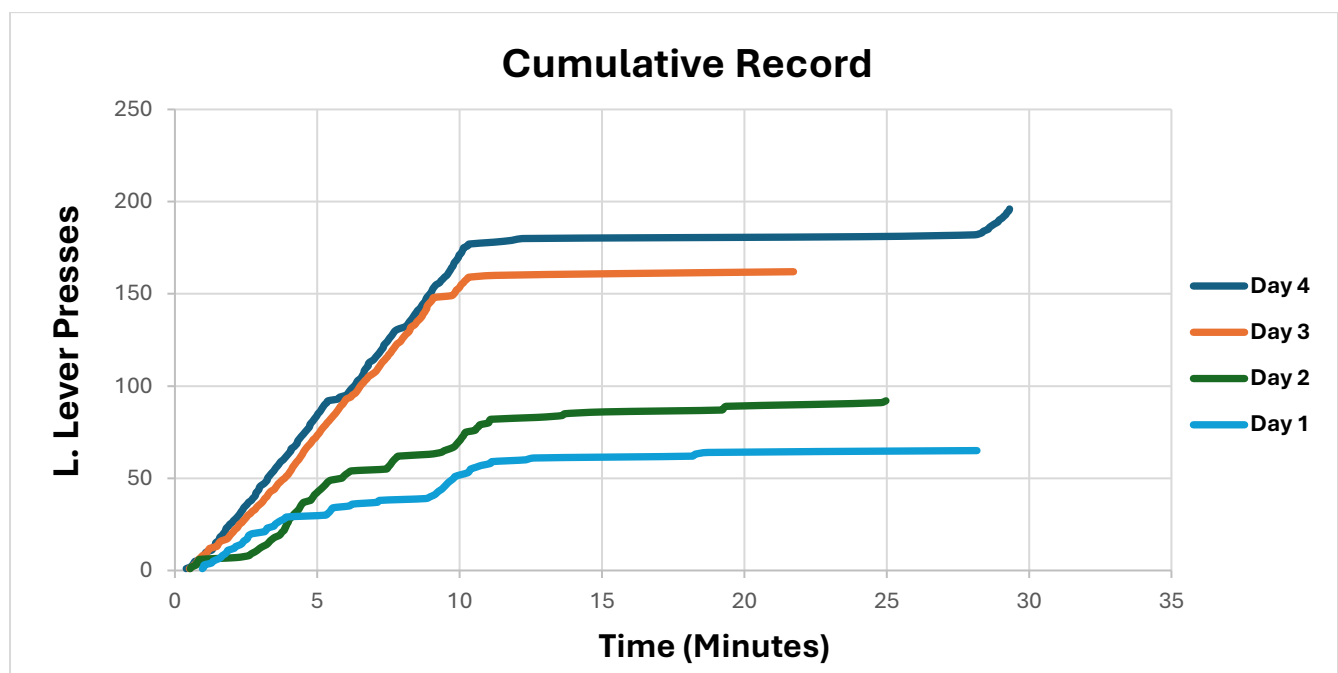
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NSC 3120 07
10/16/25

Assignment 2: Continuous Reinforcement, Extinction and Recovery

Experiment #4 graph:



Experiment #5 graph:



- In general, what does a cumulative record tell us about the rat's behavior during a training session? Give specific examples using various characteristics of the graph (*e.g. gradient*).
 - Quantifying the rat's behavior to produce a "cumulative record" graph, and plotting it against time, will ideally show a positively accelerated slope. This gradient indicates to us that with time, the rat is making more lever-presses with less time between the behaviors performed. Additionally, this indicates to us researchers that there was a successful acquisition of this new pattern of lever pressing (FR-__) which we desired to establish.
- In experiment 5, how quickly was your rat's lever-pressing behavior extinguished in part 2? That is, how many lever-presses did they perform during extinction before they ceased lever-pressing?
 - My rat's lever-pressing behavior was extinguished pretty quickly in part 2. Observing the behavioral data for day 1 and day 2, we can see that my rat made roughly 10-20 additional lever-presses after the "experimental extinction" period began (at the 10-minute mark) before fully ceasing lever-pressing behavior. Only a small amount of "spontaneous recovery" activity is seen in this graph, and it's only present around the 17-minute mark in the day 1 and day 2 slopes. The behavior was extinguished even quicker, indicated by the dramatic leveling-off in the slopes, for day 3 and day 4 with almost no additional lever-presses (less than 10) being made after the extinction period began.
- What does this result tell you about the strength of your rat's operant conditioning to lever-pressing?
 - These results tell me that the strength of my rat's operant conditioning to lever-pressing was rather weak, as my rat showed little resistance to extinction (very quick to extinguish it's lever-pressing behavior), but we can see that it's still in-tact when the rat suddenly showed spontaneous recovery around the 17-minute mark in both the day 1 and day 2 cumulative record slopes.